

**Lab Report No _12_
Understanding and practice of Timer/Counter
Control of 8051.
Embedded Systems**



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Lab Task No 1:

Solution:

Brief description (3-5 lines)
Design the Digital clock in such a way that it appears in LCM screen on 00:00:00 (hours:minutes:seconds) format. Delay Programming in msec, sec and min is further broken down into some function which is called after a specific time interval. Update screen function is invoked every 1 second to update the LCD display. 8051 Timers have been applied to achieve 1 msec accurate delay. Pressing key "1" starts the clock or resumes; pressing key "2" pauses counting and pressing key "2" again resets the clock.
The code
<pre>#include <reg51.h> #include <string.h> // LCD pin definitions sbit RS = P2^0; sbit EN = P2^1; sbit KEY1 = P1^0; // Start/Resume key sbit KEY2 = P1^1; // Pause/Reset key // Global variables for time unsigned char seconds = 0; unsigned char minutes = 0; unsigned char hours = 0; unsigned char running = 0; unsigned int ms_count = 0; // Function prototypes void init_timer0(void); void delay_ms(unsigned int ms); void lcd_cmd(unsigned char cmd); void lcd_data(unsigned char dat); void lcd_init(void); void lcd_string(unsigned char *str); void update_display(void);</pre>

```

void check_keys(void);
// Timer0 ISR
void timer0_isr(void) interrupt 1 {
    TH0 = 0xFC; // Reload timer for 1ms delay
    TL0 = 0x66;
    ms_count++;
    if(ms_count >= 1000 && running) { // 1 second has passed
        ms_count = 0;
        seconds++;
        if(seconds >= 60) {
            seconds = 0;
            minutes++;
            if(minutes >= 60) {
                minutes = 0;
                hours++;
                if(hours >= 24) {
                    hours = 0;
                }
            }
        }
        update_display();
    }
}

// Initialize Timer0
void init_timer0(void) {
    TMOD = 0x01; // Timer0 in mode 1 (16-bit)
    TH0 = 0xFC; // Initial values for 1ms delay
    TL0 = 0x66;
    ET0 = 1; // Enable Timer0 interrupt
    EA = 1; // Enable global interrupts
    TR0 = 1; // Start Timer0
}

// Millisecond delay function
void delay_ms(unsigned int ms) {
    unsigned int i;

```

```

    for(i = 0; i < ms; i++) {
        TH0 = 0xFC; // 1ms delay values
        TL0 = 0x66;
        TR0 = 1;
        while(!TF0);
        TR0 = 0;
        TF0 = 0;
    }
}
// LCD functions
void lcd_cmd(unsigned char cmd) {
    P0 = cmd;
    RS = 0;
    EN = 1;
    delay_ms(5);
    EN = 0;
}
void lcd_data(unsigned char dat) {
    P0 = dat;
    RS = 1;
    EN = 1;
    delay_ms(5);
    EN = 0;
}
void lcd_init(void) {
    lcd_cmd(0x38); // 2 lines, 5x7 matrix
    lcd_cmd(0x0C); // Display ON, cursor OFF
    lcd_cmd(0x01); // Clear display
    lcd_cmd(0x80); // Move cursor to beginning of first line
}
void lcd_string(unsigned char *str) {
    while(*str) {
        lcd_data(*str++);
    }
}
}

```

```

// Update LCD display
void update_display(void) {
    unsigned char time_str[9];
    lcd_cmd(0x85); // Position cursor at center of first line
    // Format time string
    time_str[0] = (hours/10) + '0';
    time_str[1] = (hours%10) + '0';
    time_str[2] = ':';
    time_str[3] = (minutes/10) + '0';
    time_str[4] = (minutes%10) + '0';
    time_str[5] = ':';
    time_str[6] = (seconds/10) + '0';
    time_str[7] = (seconds%10) + '0';
    time_str[8] = '\0';
    lcd_string(time_str);
}

// Check key inputs
void check_keys(void) {
    if(!KEY1) { // Start/Resume
        delay_ms(20); // Debounce
        if(!KEY1) {
            running = 1;
            while(!KEY1); // Wait for key release
        }
    }
    if(!KEY2) { // Pause/Reset
        delay_ms(20); // Debounce
        if(!KEY2) {
            if(running) { // If running, pause first
                running = 0;
            } else { // If already paused, reset
                hours = 0;
                minutes = 0;
                seconds = 0;
                update_display();
            }
        }
    }
}

```

```
        }
        while(!KEY2); // Wait for key release
    }
}
// Main function
void main(void) {
    // Initialize LCD and Timer
    lcd_init();
    init_timer0();
    // Display initial time
    update_display();
    // Main loop
    while(1) {
        check_keys();
    }
}
```

The results (Screenshot)

Build Output

```
Program Size: data=39.0 xdata=0 code=614  
creating hex file from ".\Objects\digital clock"...".  
".\Objects\digital clock" - 0 Error(s), 2 Warning(s).  
Build Time Elapsed: 00:00:00
```



Conclusion

In conclusion, we had done and performed the task practice timer/counter. In this lab task, we understood the program of digital clock by doing the 8051 Keil C language and then we implement in the flash magic by doing HEX file.